Advanced Econometrics – Exam
September 1, 2021

For all exercises use $\alpha = 5\%$ significance level.

Exercise 0 (obligatory)

Handwrite the sentence:
Dr Wozniak, you have my word I will not cheat.

Exercise 1 (30%)

Kieran O'Connor and Amar Cheema in their paper "Do Evaluations Rise With Experience?" verified the hypothesis that professors gave higher grades the longer they offered the same course. Although the paper is publicly available, the data on courses' grades is confidential. Therefore, your lecturers estimated similar models with a different dataset.

The datset for your exercise consisted of students' grades from a different university than the one of O'Connor and Cheema. Information on all courses given since Fall 2001 were collected. Some of the courses were offered since Fall 2001, some since Spring 2001, and so on. The courses differ in the length of their offerings. We removed from the database all the courses that were offered fewer than 4 times.

The table below presents results of five panel data models. Column (1) consists of fixed effects model, while columns (2)-(5) comprise random effects models' estimates. The dependent variable is *AvScore* that codes the average score of the given semester course. For a given semester students' results (on the 0-100 scale) were averaged. The set of independent variables consists of *Noffered* that assigns a consecutive number to each consecutive semester that the course is offered, *Sex* that codes the sex of a lecturer (0 for males, 1 for females), *Semester* – a dummy variable indicating a semester (0 Fall, 1 Spring), and *Year* included to account for the passage of time.

The `R` output below comes from the model with the fixed effects estimator and the Hausman test for models (1) and (2). The model had the form

$$AvScore_{it} = \beta_1 Noffered_{it} + \beta_2 Sex_{it} + u_i + \varepsilon_{it} \tag{1}$$

a) (5%) Is the panel data in the fixed effects model balanced?

b) (5%) Why was parameter for *Sex* not estimated in model (1)?

c) (5%) Interpret the Hausman test's results.

d) (5%) What can go wrong if we decided to use simple regression for this data?

e) (10%) Interpret parameters of model (4).

	Dependent variable:				
	AvScore				
	(1)	(2)	(3)	(4)	(5)
Noffered	0.499***	0.499***	0.499***	0.504***	0.476***
	(0.002)	(0.002)	(0.002)	(0.052)	(0.091)
Sex		−0.648	−0.648	−0.651	−0.313
		(0.620)	(0.621)	(0.623)	(0.983)
Semester			0.088***	0.088***	0.151***
			(0.021)	(0.021)	(0.030)
Year				−0.010	0.048
				(0.104)	(0.183)
Constant		83.424***	83.381***	104.026	−14.468
		(0.445)	(0.446)	(208.854)	(366.431)
Observations	3,131	3,131	3,131	3,131	1,504
R^2	0.967	0.967	0.967	0.967	0.967
Adjusted R^2	0.965	0.967	0.967	0.967	0.967
F Statistic	85,625.060***	45,387.610***	30,427.620***	22,813.790***	10,845.400***
Note:					*p<0.1; **p<0.05; ***p<0.01

```
Oneway (individual) effect Within Model
Call:
plm(formula = AvScore ~ as.numeric(Noffered) + Sex, data = courses,
    model = "within", index = c("Id", "Noffered"))

Unbalanced Panel: n = 200, T = 4-30, N = 3131
Residuals:
      Min.    1st Qu.     Median    3rd Qu.       Max.
-1.2738347 -0.4780633 -0.0096102  0.4804250  1.3106235
Coefficients:
                      Estimate Std. Error t-value  Pr(>|t|)
Noffered             0.4992405  0.0017061  292.62 < 2.2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
F-statistic: 85625.1 on 1 and 2930 DF, p-value: < 2.22e-16

Hausman Test
data:  AvScore ~ as.numeric(Noffered) + Sex
chisq = 0.18577, df = 1, p-value = 0.6521
alternative hypothesis: one model is inconsistent
```

Exercise 2 (30%)

A young data scientist is an intern at the Ministry of Finance. At the very beginning of her internship, she was asked to propose a simple econometric model for tax revenues (TAX). The intern started with the simple distributed lags model of the form

$$\ln{(TAX)}_t = \beta_0 + \beta_1 VAT_t + \epsilon_t, \tag{2}$$

where VAT stands for value added tax rate and is not transformed. However, for some reason she ended up with an autoregressive distributed lags model. The data available consisted of seasonally adjusted series.

```
Call:
dynlm(formula = lnTAX ~ L(lnTAX, 1) + VAT, start = 2)

Coefficients:
              Estimate Std. Error   t value  Pr(>|t|)
(Intercept)   2.039074   0.709907   2.85956     0.004 ***
L(lnTAX, 1)   0.401341   0.102283   3.96147   7.4e-05 ***
VAT           0.100613   0.010669   9.70575    <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

a) (5%) What could have been aa economic (not econometric) reason behind selection of autoregressive lags model instead of simple DL model?

b) (5%) Referring to the model results, should we keep or drop the lagged taxes from the model? Explain.

c) (10%) Are $\ln TAX$ and VAT cointegrated? Explain.

d) (5%) Let's suppose that the system is in the steady state if $VAT^* = 10$. What is the steady state value of $\ln TAX$? Explain.

e) (5%) Calculate and interpret the long-term multiplier.

Table 1: Dickey-Fuller test results

$\ln(TAX)$							
augmentations	adf	p_adf	p_bg.p.value.1	p_bg.p.value.2	p_bg.p.value.3	p_bg.p.value.4	p_bg.p.value.5
0	-3.40	0.01	0.64	0.89	0.75	0.26	0.30
1	-3.77	0.01	0.85	0.98	0.66	0.25	0.24
2	-3.08	0.03	0.92	0.99	0.71	0.18	0.18
3	-3.31	0.02	0.86	0.98	0.99	0.39	0.28

First differences of $\ln(TAX)$							
augmentations	adf	p_adf	p_bg.p.value.1	p_bg.p.value.2	p_bg.p.value.3	p_bg.p.value.4	p_bg.p.value.5
0	-9.27	0.01	0.93	0.98	0.77	0.27	0.30
1	-6.85	0.01	0.99	0.98	0.82	0.12	0.16
2	-5.01	0.02	0.84	0.96	0.99	0.16	0.19
3	-5.44	0.21	0.84	0.96	0.99	0.94	0.85

KPSS test results for VAT

```
#######################
# KPSS Unit Root Test #
#######################

Test is of type: mu with 4 lags.

Value of test-statistic is: 0.0597

Critical value for a significance level of:
                10pct  5pct 2.5pct  1pct
critical values 0.347 0.463  0.574 0.739
```

KPSS test results for ΔVAT

```
#######################
# KPSS Unit Root Test #
#######################

Test is of type: mu with 3 lags.

Value of test-statistic is: 0.0309

Critical value for a significance level of:
                10pct  5pct 2.5pct  1pct
critical values 0.347 0.463  0.574 0.739
```

Exercise 3 (40%)

"The Himalayan Database is the continuation of the work of Elizabeth Hawley, a longtime journalist based in Kathmandu, who dedicated most of her life to archiving expeditions in the Nepal Himalaya. Her detailed archives have been supplemented by information gathered from books, alpine journals and correspondence with Himalayan climbers.

The database design, computer support, and data verification effort by Salisbury has totaled more than 8000 hours during that same period. Thus, the total project time approached 20,000 hours. And this did not include the countless hours spent by Elizabeth Hawley collecting the original data during the previous 40 years!

After the initial data entry phase was completed, the American Alpine Club published the Himalayan Database in late 2004 in a booklet format with an enclosed CD-Rom containing the data and the software to access it.

[...] In 2017 the Himalayan Database was reorganized into a non-profit corporation, called The Himalayan Database. This allowed for the distribution of the database via the website and is now offered at no charge in order to increase public participation in the project."[1]

"The data cover all expeditions from 1905 through 2020 to 470 significant peaks in Nepal. Also included are expeditions to both sides of border peaks such as Everest, Cho Oyu, Makalu and Kangchenjunga as well as to some smaller border peaks. Data on expeditions to trekking peaks are included for early attempts, first ascents and major accidents.

Each expedition record contains detailed information including dates, routes, camps, use of supplemental oxygen, successes, deaths and accidents.

Each expedition record contains biographical information for all members listed on the permit as well as for hired members (e.g., Sherpas) for which there are significant events such as a summit success, death, accident or rescue."[2]

Your lecturers decided to use data from the History of the Himalayan Database to estimate an econometric model. The goal was to find factors that decide whether expedition was a success (the climbers reached the peak) or a failure. The dependent variable (`success`) and the covariates are presented in the table below. The data consists of all expeditions that took place in 2019.

Variable	Description	Values
success	Whether the climbers reached the peak	0,1
members	Number of the expedition members	$1, \ldots, 30$
hired_stuff	Number of helpers (Sherpas) hired	$0, 1, \ldots, 41$
oxygen	Use of supplemental oxygen	0,1
month	Month of the expedition	3,4,5

a) (5%) Why is this topic important? Give some economic reasons to handle the topic.

b) (5%) Are all variables in the logit model (presented below) jointly significant? Explain.

c) (10%) Interpret parameters of the logit model presented below.

d) (10%) Interpret the marginal effects.

e) (10%) Your lecturers hypothesised whether there is optimal size of the expedition. What model should be estimated to check this statement? What variable should we extend the logit model with?

[1] `https://www.himalayandatabase.com/history.html`

[2] `https://www.himalayandatabase.com/`

```
Call:
glm(formula = success ~ members + hired_staff + oxygen_used +
    month, family = binomial(link = "logit"), data = Himalayan)

Deviance Residuals:
     Min        1Q    Median        3Q       Max
-2.80936   0.06674   0.33470   0.52557   1.79568

Coefficients:
                Estimate Std. Error z value Pr(>|z|)
(Intercept)     -6.20134    2.27680  -2.724  0.00646 **
members         -0.02569    0.11146  -0.230  0.81772
hired_staff      0.17878    0.11646   1.535  0.12476
oxygen_usedTRUE  2.71190    0.54146   5.009 5.49e-07 ***
month            1.23500    0.52537   2.351  0.01874 *
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 182.59  on 168  degrees of freedom
Residual deviance: 117.76  on 164  degrees of freedom
AIC: 127.76

Number of Fisher Scoring iterations: 6

> null_logit = glm(success~1, data=Himalayan, family=binomial(link="logit"))
> lrtest(logit, null_logit)
Likelihood ratio test
Model 1: success ~ members + hired_staff + oxygen_used + month
Model 2: success ~ 1
  #Df  LogLik Df  Chisq Pr(>Chisq)
1   5 -58.881
2   1 -91.295 -4 64.828  2.798e-13 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1


Call:
logitmfx(formula = success ~ members + hired_staff + oxygen_used +
    month, data = Himalayan, atmean = TRUE)

Marginal Effects:
                     dF/dx  Std. Err.       z     P>|z|
members         -0.0029657  0.0128390 -0.2310   0.81732
hired_staff      0.0206397  0.0122419  1.6860   0.09180 .
oxygen_usedTRUE  0.4742624  0.1148157  4.1306 3.618e-05 ***
month            0.1425777  0.0637488  2.2366   0.02532 *
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
dF/dx is for discrete change for the following variables:
```